Yiyuan Lin

Research Interests: Robotics, Autonomous Systems, AI, and High-Throughput Phenotyping

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Education

Cornell University **Ithaca, NY, USA**
Ph.D. in Electrical and Computer Engineering 4.0/4.0 2023–Present (expected May 2027)
School of Electrical and Computer Engineering, Cornell Duffield Engineering
Special Committee: Dr. Yu Jiang (Chair), Dr. Kaitlin M. Gold and Dr. Nils E. Napp

Zhejiang University **Hangzhou, Zhejiang, China**
B.Eng. in Agricultural Engineering 3.9/4.0 2019–2023
College of Biosystems Engineering and Food Science & College of Control Science and Engineering
Supervisor: Dr. Fei Liu, Dr. Yanjun Cao and Dr. Fei Gao

Selected Awards and Honors

– 1st Place in the 2026 AOCABFE Student Paper Competition 2026
– 2nd Place in the 2026 AOCABFE Student Research Presentation Competition 2026
– ASABE ITSC Best Paper Award 2024 and 2026
– AG2PI Student Conference Travel Award 2024
– Zhejiang University Best Undergraduate Thesis Award 2023
– Mathematical Contest in Modeling (MCM): Finalist Award 2022
– ASABE Robotics Competition: 1st Place x2, 2nd Place x1 2021, 2022
– Chinese National Scholarship (top 1%) 2020

Experience

Cornell Communities of Practice for AI-enabled Research (CoPAIR) **Ithaca, NY**
Inaugural CoPAIR Fellow @ CoPAIR 2026–Present

IEEE-RAS Technical Committee on Agricultural Robotics and Automation (AgRA) **IEEE**
Student Representative @ IEEE-RAS TC on AgRA 2026–Present

Cornell AgriTech **Geneva, NY**
Graduate Research Assistant @ Center for Agricultural Intelligence and Robotics 2023–Present

Cornell University **Ithaca, NY**
Teaching Assistant for ECE 4110/5110 Random Signals in Communications and Signal Processing 2024–2025

Huzhou Institute of Zhejiang University **Huzhou, China**
Research Assistant at State Key Laboratory of Industrial Control and Technology @ FAST Lab 2021–2022

Grants & Proposals

Safe-Corridor, Obstacle-Aware Hybrid GNSS-Vision Navigation for Northeast Vineyards and Apple Orchards
Northeast SARE Graduate Student Research Grant (GS26-013) *Under review*
Role: Graduate Student Applicant and Project Leader **Amount requested:** $30,000
Faculty Advisor / Institutional PI: Dr. Yu Jiang 2026

Invited Talks

[1] Aerial & Ground Co-Development: Multi-Scale and Multi-Modal Disease Monitoring in Specialty Crops
NASA Acres, 2026

Publications ([†] Equal contribution; [*] Corresponding author)

[12] ***PhytoPatholoBot V2: Closing the Loop for Autonomous Vineyard Disease Scouting through 2D–3D Perception and Agentic Adaptive Revisit Planning***
Yiyuan Lin, Cole Regnier, David Combs, Lance Cadle-Davidson, Kaitlin Gold, Yu Jiang[*]
IEEE Transactions on Field Robotics (T-FR), 2026 (Under Review).

[11] ***RITE: Relation-Guided Continual Multi-Teacher Distillation for Real-Time Robotic Perception***
Yiyuan Lin, Cole Regnier, David Combs, Lance Cadle-Davidson, Kaitlin Gold, Yu Jiang[*]
IEEE Robotics and Automation Letters (RA-L), 2026 (Under Review).

[10] ***Rethinking 3D Reconstruction in the Era of Agentic AI***
Ruiming Du, **Yiyuan Lin**, Tian Qiu, Alan Zoubi, Yu Jiang[*]
HARVEST-Vision Workshop at WACV 2027 (Under Review).

[9] ***3D Grapevine Disease Quantification and Mapping***
Yiyuan Lin, Ruiming Du, Cole Regnier, Kaitlin Gold, Yu Jiang[*]
HARVEST-Vision Workshop at WACV 2027 (Under Review).
[Codebase]

[8] ***A Field-Deployable GNSS-based Navigation Stack for Outdoor Mobile Robots***
Yiyuan Lin, Cole Regnier, Yu Jiang[*]
arXiv:2609.28933, 2026.
[Preprint] [Codebase]

[7] ***SAM-CLIP-Thermal: Leveraging Large Multimodal Models for Reliable and Scalable Annotation in Thermal Image Segmentation for Field Plant Phenotyping***
Yiyuan Lin[†], Chenjiao Tan[†], Changying Li[*], Yu Jiang[*]
Plant Phenomics, 8(4), 100264, 2026.
[Paper] [Codebase] [Dataset] [Project Page]

[6] ***Robot-Enabled Field Phenotyping of Grape Berry Cluster Color Using Multimodal Vision Foundation Models for Genetic Mapping***
Yiyuan Lin, Madan Pandey, Lance Cadle-Davidson, Matthew Clark, Soon Li Teh, Yu Jiang[*]
ASABE Annual International Meeting, Paper No. 2600399, 2026.
– **1st Place in the 2026 AOCABFE Student Paper Competition**
– **2nd Place in the 2026 AOCABFE Student Research Presentation Competition**
[Paper] [Codebase] [Project Page]

[5] ***Beyond Seasonality: A High-Fidelity Apple Orchard Digital Twin***
Ruiming Du, **Yiyuan Lin**, Daoyuan Jin, Nicholas Gunner, Yu Jiang[*]
ASABE Annual International Meeting, Paper No. 2600941, 2026.
– **2026 ITSC Best Paper Award**
[Paper] [Project Page]

[4] ***Integrating Large Multi-Modal Models for Automated Powdery Mildew Phenotyping in Grapevines***
Yiyuan Lin, Zachary Dashner, Ana Jimenez, Dustin Wilkerson, Lance Cadle-Davidson, Summaira Riaz[*], Yu Jiang[*]
Artificial Intelligence in Agriculture, 2026.
[Paper] [Codebase] [Dataset] [Project Page]

[3] ***DATR: Diffusion-Based 3D Apple Tree Reconstruction Framework With Sparse-View***
Tian Qiu, Alan Zoubi, **Yiyuan Lin**, Ruiming Du, Lailiang Cheng, Yu Jiang[*]
IEEE Robotics and Automation Letters (RA-L), 11(3), 2810–2817, 2026.
[Paper] [Codebase] [Project Page]

[2] ***Effective integration of vision foundational models for semantic segmentation to quantify grape foliage powdery mildew infection***
Yiyuan Lin, Anna Underhill, Lance Cadle-Davidson, Ana Jimenez, Summaira Riaz, Yu Jiang*
ASABE Annual International Meeting, Paper No. 2401108, 2024.
– **2024 ITSC Best Paper Award**
[Paper] [Codebase] [Dataset] [Project Page]

[1] ***Automated rice phenology stage mapping using UAV images and deep learning***
Xiangyu Lu, Jun Zhou, Rui Yang, Zhiyan Yan, **Yiyuan Lin**, Jie Jiao, Fei Liu*
Drones, 7(2), 83, 2023.
[Paper] [Project Page]

Outreach and Extension

More than 50 field robot demonstrations and extension events (2024–2026), with an estimated cumulative attendance of over 1,000. Selected engagements:

[5] **VitisGen3 and Vitis x Muscadinia Project Teams** 2026

[4] **Cornell Fruit Field Day** 2026

[3] **U.S. Department of Agriculture (USDA)** 2025

[2] **NASA Acres Consortium** 2024

[1] **Food and Agriculture Organization of the United Nations (FAO)** 2024

Selected Robotic Projects

PPB-OTR-UVC [Codebase] [Project Page] 2026
– Integrated dual-GNSS autonomous navigation with prescription-map-based UV-C lamp control for targeted disease management using spatial disease data from PPBv2 or PPB-NG.
– Coordinated treatment and transit speeds to deliver target UV doses within prescribed zones and reduce travel time between zones.
– Collaborated with plant pathologists to evaluate the effectiveness of robotic UV-C treatments for plant disease management.

PhytoPatholoBot Next Generation (PPB-NG) [Codebase] [Project Page] 2025
– Developed an autonomous robotic sensing platform combining RGB, thermal (3-5 µm), VNIR (400–1000 nm), and SWIR (1000–2500 nm) imaging for high-throughput field phenotyping.
– Integrated image acquisition with GNSS positioning, heading, attitude, and timing metadata for spatial referencing and cross-modal data alignment.
– Collaborated with breeders on grape berry Brix estimation and with plant pathologists, breeders, and geneticists on disease phenotyping.

PhytoPatholoBot v2 (PPBv2) [Codebase] [Project Page] 2024
– Developed an autonomous robotic phenotyping robot with synchronized stereo RGB imaging and active strobe illumination for consistent daytime canopy observations.
– Integrated dual-GNSS autonomous navigation and GPS-synchronized data logging for centimeter-level localization and image-to-plant spatial mapping.
– Collaborated with plant pathologists, breeders, and geneticists on disease phenotyping, berry size and color measurement, and budbreak timing.

Academic Services

- **Inaugural Fellow:** Cornell Communities of Practice for AI-enabled Research (CoPAIR)
- **Student Representative:** IEEE RAS Technical Committee on Agricultural Robotics and Automation (AgRA)
- **Journal Reviewer:** IEEE Robotics and Automation Letters (RA-L); Plant Phenomics; Tropical Plant Pathology; Applied Engineering in Agriculture; Journal of the ASABE, etc.
- **Conference Reviewer:** ICRA; IROS; etc.

Mentorship

Apurva Singh, M.Eng. student at Cornell University 2026–Present
- **Topic:** GNSS–vision hybrid navigation for field robots

Yukta Karki, Undergraduate at University of Louisiana Monroe 2026
- **Topic:** Vision-based field robot navigation

Owen Bates, Undergraduate at Rochester Institute of Technology 2024–2025
- **Topics:** Autonomous robotic swarm navigation (2024); robotic solutions for vineyard management (2025)

Mridula Buragohain, Undergraduate at Indian Institute of Technology Kharagpur 2024
- **Topics:** LLMs and vision foundation models for grape disease phenotyping; robotic arm motion planning
- **Next step:** Ph.D. program, Arizona State University

Albert Sun, Undergraduate at Cornell University 2024
- **Topic:** Autonomous robotic swarm navigation
- **Next step:** Apex, Los Angeles, CA, USA